STANDUP COMEDY TOOLKIT

YOUR COMEDY OPERATING SYSTEM · TECHNICAL PRD v1.0

Document Version	1.0 — Initial Release
Prepared For	Engineering, Design & QA Teams
Platform	React Native (Expo) — iOS & Android
Storage Strategy	Offline-First (AsyncStorage + Expo SQLite) — Cloud-ready for Phase 2
Future Backend	Supabase PostgreSQL + Auth — Phase 2 roadmap
Design Language	Dark OLED (#1C1C28) · Gold (#FFB800) · Purple (#7C3AED) — see Section UI
Target Users	Stand-up comedians: beginners through professionals
Confidentiality	Internal Use Only
Year	2026

This document is the single source of truth for all engineering, design, and QA decisions. All team members must read and acknowledge this PRD before commencing any implementation.

UI. APP SCREEN DESIGN REFERENCE

The four screens below define the approved visual language of the StandUp Comedy Toolkit. The design is built on a **deep OLED dark theme** (#1C1C28 background) with **amber gold** (#FFB800) as the primary creative accent and **deep purple** (#7C3AED) for secondary CTAs. The notepad metaphor in the Joke Builder uses a warm cream (#FFF8E1) surface with ruled lines to evoke a physical comedy notebook. All screens must match these specs exactly.



Figure 1: Approved design reference — Login · Dashboard (Home) · Joke Builder · Setlist Builder

Screen-by-Screen Design Specifications

Element	Specification	Screen(s)
App Background	#1C1C28 — deep charcoal OLED; maximises battery life on OLED panels	All
Card Surface	#252535 — slightly lighter than bg; 12dp radius; subtle #3D3D4E border	All cards
Primary Gold	#FFB800 — key actions, highlights, brand text, active nav items	All
Gold Light	#FFD700 — hover states, donut arc fill, mic sound lines	Splash, Home
Primary Purple	#7C3AED — secondary CTAs, punchline label, active tab indicator	Joke Builder
Soft Purple	#9B72CF — login button, Add to Setlist, Practice Set buttons	Login, Builder
Notepad Surface	#FFF8E1 warm cream — Joke Builder notepad card; ruled lines #E5DCC8	Joke Builder
Spiral Binding	#C0B080 circle outlines left edge of notepad — evokes physical notebook	Joke Builder
Typography	Inter variable: Bold 22sp screen titles, SemiBold 16sp labels, Regular 14sp body	All
Status Bar	9:41 timestamp white left; signal icons white right; transparent overlap on dark bg	All
CTA Buttons	Full-width, 24dp radius, 48dp height min. Gold fill for primary; purple fill for secondary	All

Element	Specification	Screen(s)
Bottom Nav	#151520 bar; active = gold icon + label; inactive = #5B5B6E; center '+' = purple bubble	Home, Setlist
Dashed Border Box	#4A4A5A dashed outline — Add Joke slot in Setlist Builder	Setlist Builder
Progress Bars Chart	Alternating gold / purple bars; M-I day labels in muted gray	Home Dashboard
Tag Chips	#E5DCC8 cream bg; #555544 text; 10dp radius — in Joke Builder notepad	Joke Builder
Time Badge	#FFB800 gold pill — set duration in Setlist Builder top-left	Setlist Builder

1. PROJECT VISION & PHILOSOPHY

The StandUp Comedy Toolkit is a mobile-first, offline-first **Comedy Operating System** designed exclusively for stand-up comedians. It solves a real, documented problem: comedians currently manage their material across fragmented tools — voice memos, scattered notes apps, physical notebooks — with no unified system connecting the journey from raw idea to polished stage performance.

> Core Vision: From Idea to Stage — One System Most comedy apps do one thing: store jokes, time sets, or record performances. The StandUp Comedy Toolkit does all of it in a single structured workflow: Idea Captured → Joke Structured → Set Ordered → Practice Timed → Performance Recorded → Joke Rated → Material Improved

1.1 Problem Statement

- Joke ideas get lost across voice memos, notes apps, and physical notebooks — no unified capture system.
- Comedians have no structured way to move from raw premise to polished joke with Setup, Punchline, Tag, and Callback.
- Tracking which jokes land with which crowd types is manual and inconsistent.
- Rewriting and iterating on material is chaotic — no version history means losing what worked before.
- No existing app combines writing, performing, and improvement in one coherent flow.
- Even professional comedians on Reddit report still relying on generic notes apps.

1.2 Target Users

User Type	Description	Primary Need
Beginner comedian	Open-mic regulars; 0–2 years of stage time	Idea capture + structured joke writing
Developing comedian	Regular performer; building a tight 10-minute set	Setlist curation + performance tracking
Semi-professional	Paid gigs; adapts material for different crowds	Crowd-type adaptation + version control
Comedy writer	Writes for others; needs organisation system	Premise expansion + tag system + export
Storytelling comedian	Long-form personal narrative style	Character builder + script teleprompter

1.3 Color Palette — App UI

Token	Hex	Usage	Screen
OLED Black	#1C1C28	App background — OLED battery saving	All
Card Dark	#252535	Card surfaces, modals, input fields	All cards
Card Darker	#333345	Inner card elements, tag bg, progress bg	Home, Builder
Primary Gold	#FFB800	Brand, active nav, highlights, primary CTA	All
Gold Light	#FFD700	Sound lines, secondary gold accents	Splash
Primary Purple	#7C3AED	Punchline label, active indicators	Joke Builder
Soft Purple	#9B72CF	Secondary buttons, setlist CTA	Login, Setlist
Notepad Cream	#FFF8E1	Joke Builder notepad surface	Joke Builder
Notepad Lines	#E5DCC8	Ruled lines on notepad card	Joke Builder
Text White	#FFFFFF	Primary text on dark backgrounds	All
Text Gray	#8B8B9E	Secondary labels, subtitles, metadata	All
Text Muted	#5B5B6E	Inactive nav, timestamps, captions	All

Token	Hex	Usage	Screen
Status Raw	#EF4444	Idea status: Raw	Idea Inbox
Status Dev	#F59E0B	Idea status: Developing	Idea Inbox
Status Stage	#10B981	Idea status: Stage Ready	Idea Inbox

2. PRODUCT FEATURES & SCREEN INVENTORY

The app ships with 12 named feature modules across three phases. Phase 1 (MVP) is strictly offline — no account required, no internet. Phase 2 introduces cloud sync and AI features. Phase 3 adds social and live-performance integrations.

2.1 Full Feature & Screen Inventory

#	Screen / Feature	Phase	Key Components	Business Rule
1	Idea Inbox	MVP	Voice/text capture, status badge, search, sort	Ideas have 3 statuses: Raw / Developing / Stage Ready
2	Joke Builder	MVP	Notepad UI, Setup/Punchline/Tag/Callback fields, save	All 4 joke structure fields optional except Setup
3	Setlist Builder	MVP	Drag-to-reorder, time per joke, total time badge	Total time = sum of per-joke durations; max 60 min
4	Practice Mode	MVP	Countdown timer, large teleprompter text, stage sim	Timer counts from total set time to zero
5	Home Dashboard	MVP	Goal widget, Idea Inbox shortcut, Quick Actions, chart	Today's Goal = daily write + practice targets
6	Login / Onboarding	MVP	Email+password, social auth placeholders, Create Acct	Phase 1: local auth only; Phase 2: Supabase Auth
7	Premise Expander	v1.1	AI angle generator, premise input, output card list	Phase 2 only — Supabase Edge Fn + OpenAI
8	Performance Recorder	v1.1	Audio/video record, local storage, playback	Stored on-device; Phase 2: Supabase Storage upload
9	Joke Rating System	v1.1	Post-show rating 1–10, laugh score, notes	One rating per joke per performance date
10	Version Control	v1.1	Joke v1/v2/v3 timeline, diff view, revert	Max 20 versions per joke; oldest pruned first
11	Laugh Analyzer	v2.0	Audio upload, laughter intensity detection, heatmap	Requires OpenAI Whisper or dedicated audio ML model
12	Open Mic Finder	v2.0	Map view, venue list, filters, RSVP link	Requires location permission + third-party events API
13	Topic Explorer	v2.0	Writing prompt generator, category filters	25 built-in prompt categories; AI expansion in Phase 2
14	Character Builder	v2.0	Character name, traits, voice, sample lines	Linked to jokes for storytelling comedians
15	Profile & Settings	v1.1	Comedy style, crowd preference, export, notifications	Style affects AI tone; crowd affects AI suggestions

2.2 MVP Acceptance Criteria

Feature	Description	Acceptance Criteria
Idea Inbox	Capture raw comedy ideas instantly by text or voice	Idea saved with timestamp; status defaults to 'Raw'; searchable
Joke Builder	Structure joke with Setup, Punchline, Tag, Callback	All fields save to SQLite; joke linked to parent idea if created from Inbox
Setlist Builder	Drag jokes into ordered set with per-joke timing	Drag reorder persists; total time updates in real time; max 60 min enforced

Feature	Description	Acceptance Criteria
Practice Mode	Stage simulation with timer and teleprompter view	Timer starts/pauses/resets; large text; shows current joke name in set
Home Dashboard	Overview of today's goals and quick actions	Goal progress updates live; Inbox badge shows unreviewed idea count
Offline Persist.	All data survives app close and restart	Data intact after force-kill and relaunch; no data loss

3. IDEA INBOX — BUSINESS RULES & WORKFLOWS

The Idea Inbox is the fastest-access screen in the app. When inspiration strikes on the bus, in the shower, or watching the news — the user taps once and starts capturing. Speed of capture is the single most critical UX metric for this screen.

> Design Principle: 'One Tap to Capture' The Idea Inbox must be reachable in ≤2 taps from anywhere in the app. The floating '+' button in the bottom nav opens the capture sheet instantly. The microphone and keyboard modes must load in under 300ms.

3.1 Idea Status Workflow

Status	Color	Meaning	Transitions
Raw	#EF4444 Red	Just captured; unrefined thought	→ Developing when user starts editing or adds punchline
Developing	#F59E0B Amber	Being actively worked on	→ Stage Ready when joke is marked complete by user
Stage Ready	#10B981 Green	Polished; approved for setlist use	→ Developing if user returns to edit after performance feedback
Archived	#5B5B6E Gray	Old idea no longer being developed	Archived ideas hidden by default; accessible via filter

3.2 Capture Modes

- **Text mode:** Full-screen text input with auto-save every 2 seconds. No 'Save' button required — the idea is always persisted.
- **Voice mode:** Activates device microphone; records audio clip stored locally (m4a format); transcript generated on-device via Apple Speech/Android SpeechRecognizer in Phase 1.
- **Quick capture:** Widget accessible from home screen (iOS/Android) — opens directly to blank text input bypassing the app launch flow.
- **Share sheet integration (v1.1):** 'Share to StandUp' appears in the iOS/Android share sheet — text from any other app can be sent directly to Idea Inbox.

3.3 Idea Fields

Field	Type	Description	Validation
content	TEXT	The raw idea text — premise, observation, phrasing	Required; 1–2000 chars
title	TEXT	Auto-generated from first 50 chars of content	Optional; editable
status	ENUM	Raw / Developing / Stage Ready / Archived	Default: Raw
voice_clip	TEXT path	Local file path to .m4a voice recording if captured	Optional; nullable
transcript	TEXT	Auto-transcribed text from voice clip	Optional; nullable
tags	TEXT array	User-defined topic tags for search and filtering	0–10 tags; each max 30 chars
comedy_style	ENUM	Observational / Dark / Storytelling / Roast	Inherits from profile default
created_at	DATETIME	Timestamp of capture	Auto-set; immutable
updated_at	DATETIME	Last edit timestamp	Auto-updated on every save

4. JOKE BUILDER — STRUCTURE, BUSINESS RULES & VALIDATION

The Joke Builder is the creative core of the app. It teaches and enforces the fundamental structure of stand-up comedy writing: **Setup** → **Punchline** → **Tag** → **Callback**. The notepad UI deliberately evokes a physical comedy notebook to feel familiar and creatively inviting.

4.1 Joke Structure Definitions

Component	Definition	Example
Setup	The premise or question that creates expectation	'Dating apps are weird...'
Punchline	The twist or subversion that creates the laugh	'It's like ordering humans on Swiggy.'
Tag	A second, shorter punchline that rides the first laugh	'And it has a rating system. She's a 4.2.'
Callback	A later reference back to an earlier joke in the set	'Remember that Swiggy delivery? He's still on the way.'
Bit	A complete joke unit: Setup + 1 or more Punchlines	Any of the above combined into one stage segment
Crowd Work	Spontaneous audience interaction — logged as a note	'Asked front row where they were from. Got 5 minutes from it.'

4.2 Joke Builder Validation Rules

- **Setup field:** Required; minimum 5 characters; maximum 500 characters.
- **Punchline field:** Optional in draft status; required before joke can be marked 'Stage Ready'.
- **Tag field:** Optional; if present, minimum 5 characters; maximum 300 characters.
- **Callback field:** Optional; maximum 300 characters.
- **Tags (topic labels):** 0–10 tags; each tag maximum 30 characters; auto-suggest from existing user tags.
- **Estimated time:** Required before joke can be added to Setlist; entered in 0.5-minute increments.
- **Comedy style:** Defaults to profile setting; overridable per joke.
- **Crowd type:** Optional — Corporate / College / Club / Any; affects AI suggestions in Phase 2.
- **Auto-save:** Joke content saves every 3 seconds of inactivity — no explicit Save action required.

4.3 Joke Lifecycle States

State	Badge Color	Meaning	Entry Condition
Draft	Gray	Being written; incomplete structure	Default on creation
Developing	Amber	Has Setup + Punchline; being refined	Auto when punchline saved
Stage Ready	Green	Polished; approved for setlist inclusion	User manually promotes
Retired	Red	Joke removed from active rotation	User retires after poor ratings
In Set	Gold	Currently assigned to an active setlist	Auto when added to any setlist

4.4 Premise Expander — AI Feature (Phase 2)

When a user has a premise but is stuck finding angles, the Premise Expander calls a Supabase Edge Function that prompts GPT-4o-mini with the user's premise, comedy style, and crowd type to generate 5–8 alternative angles or unexpected punchline directions.

User Premise	Comedy Style	AI Generated Angles
Dating apps	Observational	Height filter lies · Bio hobby fraud · 'Looking for something casual' meaning · Superlike cringe · Unmatch
Indian parents	Storytelling	Degree as social currency · Marriage timeline pressure · WhatsApp forward as love language · 'Log kya
Gym first week	Self-deprecating	Resolution crowd vanishes by Feb 10 · Equipment tutorials nobody read · Mirror selfie etiquette · Cardio

5. SETLIST BUILDER & PRACTICE MODE

The Setlist Builder transforms individual jokes into a structured, timed performance. It is the bridge between writing and the stage. The Practice Mode then simulates the live performance experience with a countdown timer and teleprompter view.

5.1 Setlist Builder Business Rules

- **Setlist creation:** A setlist must have a name (required, 2–60 chars) and a target total duration.
- **Adding jokes:** Only jokes in 'Stage Ready' state can be added to a setlist. Draft jokes are excluded.
- **Ordering:** Jokes are displayed in a numbered list; drag-to-reorder updates order_index. Order persists across app restarts.
- **Per-joke timing:** Each joke has an estimated_duration in 0.5-minute increments. Required before setlist can be practised.
- **Total time badge:** Computed in real-time = SUM of all joke durations. Displayed in gold pill top-left.
- **Time cap:** Maximum 60 minutes per setlist. Adding a joke that would exceed 60 min shows a warning.
- **Multiple setlists:** Users can have unlimited saved setlists. Active setlist is pinned to Practice Mode.
- **Setlist duplication:** Long-press → Duplicate creates a copy with '(copy)' suffix for crowd-type variants.
- **Crowd adaptation tag:** Each setlist can be tagged Corporate / College / Club — used for AI suggestions in Phase 2.

5.2 Setlist Data Schema

Field	Type	Description	Notes
id	UUID	Primary key	Client-generated
name	TEXT	Setlist name, e.g. '10-min open mic set'	Required; unique per user
jokes	JSON	Ordered array of {joke_id, order_index, duration}	Recomputed on every reorder
total_duration	REAL	Sum of all joke durations in minutes	Computed; stored for display speed
crowd_type	ENUM	Corporate / College / Club / Any	Default: Any
is_active	BOOL	The setlist currently selected for Practice Mode	Only one can be active at a time
last_performed	DATE	Date of most recent performance using this set	Nullable; set by user
created_at	DATETIME	Creation timestamp	Immutable

5.3 Practice Mode — Behaviour

Feature	Specification
Timer	Counts down from total setlist duration to zero. Visual ring indicator at 50%, 25%, and 0% thresholds
Teleprompter view	Large 22sp text; shows Setup of current joke; auto-advances on single tap. Background stays dark
Joke name display	Bold gold text at top showing current joke name and position e.g. '2 / 4 — Dating Apps'
Pause / Resume	Single tap on timer pauses; second tap resumes. Timer state persists through screen-off
Skip / Back	Swipe right to skip to next joke; swipe left to go back. Updates timer position
Session log	On practice end, prompts user to rate each joke 1–5 stars. Saved to local performance log
Stage ambient mode	Optional: dims all UI except joke text and timer for distraction-free presentation
Finish screen	Shows total time used vs target, per-joke time breakdown, prompt to log as a performance

6. PERFORMANCE TRACKING — RECORDER, RATINGS & VERSION CONTROL

Professional improvement in stand-up requires honest feedback loops. These three features close the loop between stage performance and written material — forming the 'improve' cycle of the Comedy OS.

6.1 Performance Recorder

- **Recording modes:** Audio-only (default) and video (opt-in; requires camera permission).
- **Storage:** Recordings saved locally as .m4a (audio) or .mp4 (video) in the app's sandboxed documents directory.
- **File naming:** Auto-named with format: setlist_name_YYYY-MM-DD_HH-MM.m4a
- **Max duration:** 90 minutes per recording. Warning shown at 80 minutes.
- **Transcript (v1.1):** On-device speech-to-text using Apple Speech Framework (iOS) or Android SpeechRecognizer. Transcript stored alongside recording.
- **Playback:** Built-in player with waveform visualisation; jump to timestamp by tapping transcript lines.
- **Phase 2 — Cloud upload:** Option to upload to Supabase Storage for cross-device access and backup.

6.2 Joke Rating System

After each performance session, the user rates each joke they performed. Ratings are stored per-performance-date, not overwritten — historical ratings are always preserved.

Field	Type	Description	Notes
joke_id	UUID	Reference to joke in jokes table	Foreign key
performance_date	DATE	Date of the open mic or practice session	Required
laugh_score	INT 1–10	User-assigned audience reaction rating	1=silence, 10=huge laugh
crowd_type	ENUM	Crowd that night: Corporate/College/Club/Other	Affects improvement suggestions
notes	TEXT	Freeform post-show notes about the joke	Optional; max 500 chars
kept	BOOL	Did user keep this joke in rotation after show?	Default: true

Average laugh score is computed across all performances and displayed as a badge on the joke card. Jokes with average < 4.0 after 3+ performances receive a 'Low Performer' flag with a suggestion to rewrite.

6.3 Version Control — Joke Revisions

- **Auto-versioning:** Every time a joke's Setup, Punchline, Tag, or Callback field is edited and saved, the previous state is snapshotted as a new version row.
- **Version cap:** Maximum 20 stored versions per joke. When exceeded, the oldest version (lowest version_number) is purged.
- **Version metadata:** Each version stores: version_number (INT), content_snapshot (JSON of all fields), created_at, edit_note (optional user comment).
- **Revert:** User can tap any version and choose 'Revert to this version' — the current joke fields are overwritten with the snapshot. A new version is created for the revert action itself.
- **Diff view (v1.1):** Side-by-side comparison of any two versions highlighting changed text in gold.
- **Version timeline:** Displayed as a horizontal scrollable timeline with version number and date.

7. DATA ARCHITECTURE — OFFLINE-FIRST STORAGE

Phase 1 is **strictly offline**. All user data lives on-device using **AsyncStorage** (key-value preferences and metadata) and **Expo SQLite** (relational data for ideas, jokes, setlists, ratings, and versions). The schema is designed to map directly to Supabase PostgreSQL in Phase 2 with no migration required — identical table structure, same UUID strategy.

7.1 Storage Strategy

Data Type	Storage	Key / Table	Rationale
User profile	AsyncStorage	@su_user_profile	JSON object; comedy style, crowd pref, display name
App settings	AsyncStorage	@su_settings	Theme, notification prefs, default crowd type
Auth token (P2)	SecureStore	@su_auth_token	JWT stored in Keychain/Keystore — Phase 2 only
Ideas	Expo SQLite	ideas	High create volume; status, voice path, transcript, tags
Jokes	Expo SQLite	jokes	Core content table; setup/punchline/tag/callback fields
Joke versions	Expo SQLite	joke_versions	Append-only snapshot table; max 20 per joke
Setlists	Expo SQLite	setlists	Ordered joke list with per-joke timing
Performances	Expo SQLite	performances	Performance sessions with setlist reference and date
Joke ratings	Expo SQLite	joke_ratings	One row per joke per performance date; laugh score
Voice recordings	File system	documents/recordings/	Binary .m4a / .mp4; path stored in ideas/performances tables
Goal progress	AsyncStorage	@su_daily_goals	Today's write count + practice minutes; reset at midnight

7.2 SQLite Table Definitions

Table	PK	Key Columns	Notes
ideas	id (UUID)	user_id, content, title, status (ENUM), voice_clip, transcript, tags (JSON), comedy_style, created_at, updated_at	status enum: raw/developing/stage_ready/archived
jokes	id (UUID)	user_id, idea_id, setup, punchline, tag, callback, est_duration REAL, state (ENUM), comedy_style, crowd_type, avg_score REAL, created_at, updated_at	idea_id nullable for jokes created directly
joke_versions	id (UUID)	joke_id, version_number INT, content_snapshot JSON, edit_note TEXT, created_at	Max 20 rows per joke_id; oldest purged
setlists	id (UUID)	user_id, name, jokes JSON, total_duration REAL, crowd_type, is_active BOOL, last_performed DATE, created_at	jokes: [{joke_id, order_index, duration}]
performances	id (UUID)	user_id, setlist_id, performed_date, recording_path TEXT, transcript TEXT, total_time_used REAL, notes TEXT	One row per performance session
joke_ratings	id (UUID)	joke_id, performance_id, laugh_score INT, crowd_type, notes TEXT, kept BOOL	Unique on (joke_id, performance_id)

7.3 Data Integrity Rules

- **Cascade archive:** Archiving an idea does not delete linked jokes. Jokes inherit is_archived=true but can be promoted independently.
- **Version cap enforcement:** On every joke save, the version count is checked. If count ≥ 20, DELETE WHERE id = (SELECT id FROM joke_versions WHERE joke_id = ? ORDER BY version_number ASC LIMIT 1).

- **Setlist total time:** Recomputed on every joke add/remove/duration-edit. Stored for display performance but always recomputable.
- **Midnight goal reset:** @su_daily_goals AsyncStorage entry is cleared at device midnight. New day starts fresh.
- **Recording file integrity:** On app launch, orphaned recording_path references (where file no longer exists) are set to NULL.

8. PERSONALIZATION ENGINE & COMEDY STYLE SYSTEM

The personalization system connects every feature to the user's stated comedy identity. In Phase 1, personalization drives content organisation and display. In Phase 2, it powers AI-generated suggestions that match the user's unique voice.

8.1 Comedy Style Profiles

Style	Description	AI Tone (Phase 2)
Observational	Finds humour in everyday situations, social norms, and shared experiences	Relatable, audience-inclusive, 'we've all been there' framing
Dark Humour	Explores taboo topics, absurdist logic, and uncomfortable truths	Subversive, unexpected, challenges assumptions
Storytelling	Personal narrative arcs with character development and callbacks	Chronological setup, escalating tension, satisfying punchline
Roast / Topical	Punches at public figures, current events, or audience members	Sharp, topical references, contemporary cultural hooks
Self-deprecating	Vulnerability-based humour; the comedian is the target	First-person tone; honest; disarming opener style
Absurdist	Surreal logic, unexpected leaps, deliberately illogical conclusions	Non-sequitur friendly; unusual perspective framing

8.2 Crowd Type Adaptation

Every joke and setlist can be tagged with a target crowd type. In Phase 2, the AI uses the crowd type to suggest adaptations — e.g., toning down an edgy punchline for a corporate crowd or adding more risqué angles for a late-night club audience.

Crowd Type	Characteristics	AI Adaptation Guidance
Corporate	Mixed age, formal setting, low risk tolerance	Remove profanity; avoid divisive topics; keep relatable, universal
College	18–25, high energy, current references expected	Pop culture hooks; meme-aware; self-aware meta-humour works well
Club	Late night, drinking, high risk tolerance	Edgier punchlines acceptable; crowd work encouraged; higher energy
Any / Open	Unknown mix — safest default for new material	Balanced tone; test setup first; observe before escalating edge

8.3 Punchline Finder AI — Phase 2

The Punchline Finder is one of the app's signature AI features. The user writes the Setup of a joke and taps 'Find Punchlines'. A Supabase Edge Function sends the setup, comedy style, and crowd type to GPT-4o-mini and returns 4–6 candidate punchlines ranked by estimated comedic impact.

User Setup	Style	Generated Punchlines (ranked)
My gym trainer said 'no pain, no gain'	Self-dep.	1. I said great, because I've been avoiding both. 2. I nodded and went home. 3. So I gave him
Indian parents treat degrees like Pokémon	Observational	1. Gotta catch them all before marriage season. 2. 'Can't evolve without an MBA.' 3. The rares

9. TECHNOLOGY STACK — PHASE 1 (OFFLINE) & PHASE 2 (CLOUD)

The same proven stack as previous projects — chosen for developer velocity, zero infrastructure cost in Phase 1, and clean migration to cloud in Phase 2. Every Phase 1 technology maps directly to a Phase 2 equivalent.

9.1 Phase 1 — Offline Stack (MVP)

Layer	Technology	Version	Purpose
App Framework	React Native	0.74.x	Cross-platform iOS & Android — same codebase
Dev Toolchain	Expo SDK	51.x	Managed workflow; EAS Build; OTA JS updates
Language	TypeScript	5.4.x	Strict mode; full type safety across codebase
Navigation	Expo Router (file-based)	3.5.x	Stack + Tab + Modal; typed route params
State	Zustand	4.5.x	Global store for jokes, setlists, goals; no boilerplate
Local DB	Expo SQLite	13.x	Relational storage for ideas, jokes, setlists, ratings
Simple Storage	AsyncStorage	1.23.x	User profile, settings, daily goal cache
Drag & Drop	React Native Reorderable List	0.8.x	Setlist Builder drag-to-reorder with haptic feedback
Audio Record	Expo AV	14.x	Voice capture in Idea Inbox and Performance Recorder
Speech-to-Text	Expo Speech (on-device)	12.x	Voice transcript in Idea Inbox — no cloud call
Forms	React Hook Form + Zod	7.x/3.x	Joke Builder, Idea form — validation + schema
Animations	React Native Reanimated	3.x	Timer ring, card entrance, drag feedback
Icons	@expo/vector-icons (Feather)	14.x	Consistent icon set; gold tinted for active states
Fonts	Expo Google Fonts (Inter)	12.x	Inter variable — all typography
Haptics	Expo Haptics	13.x	Drag reorder, checkbox toggle, timer end
Notifications	Expo Notifications (local)	0.28.x	Daily write reminder, practice nudge
Date Utilities	date-fns	3.x	Performance date formatting, goal window calc
Testing	Jest + RNTL	29.x	Unit tests; SQLite mocked with in-memory DB
E2E Testing	Maestro	1.x	Full flow: idea → joke → setlist → practice

9.2 Phase 2 — Cloud Stack (Future)

Layer	Technology	Replaces / Extends	Notes
Auth	Supabase Auth	Local profile in AsyncStorage	Email/Password + Google + Apple OAuth
Cloud DB	Supabase PostgreSQL 15	Cloud mirror of local SQLite	Same schema; offline data synced on first login
Sync Engine	Zustand + custom middleware	New addition	Write-local-first; push pending to Supabase on connectivity
AI — Punchlines	Supabase Edge Fn + OpenAI	No AI in Phase 1	GPT-4o-mini; comedy style + crowd type in prompt
AI — Expander	Supabase Edge Fn + OpenAI	No AI in Phase 1	5–8 premise angles; ranked by comedic potential
AI — Analyzer	Whisper API / custom ML	No analyzer in Phase 1	Laughter detection from audio recording
File Storage	Supabase Storage	Local file system only in P1	Recordings uploaded optionally for cloud backup

Layer	Technology	Replaces / Extends	Notes
Open Mic API	Eventbrite / Meetup API	No map feature in Phase 1	Location-based open mic discovery
Analytics	PostHog	None in Phase 1	Privacy-first; no PII; funnel analysis

10. SECURITY, PRIVACY & DATA GOVERNANCE

Phase 1 makes zero network requests — the entire app operates locally. The data stored includes personal creative material (jokes, ideas, voice recordings) that comedians may consider highly private or commercially valuable. All protections are implemented from day one.

10.1 Data Sensitivity Classification

Data Field	Level	Storage	Encrypted	Notes
Jokes / Scripts	High	SQLite	AES-256	Original creative material — commercially sensitive
Voice recordings	High	File system	AES-256	Contains vocal identity; subject to platform-level encryption
Transcripts	High	SQLite	AES-256	Derived from voice; treated with same sensitivity
Performance notes	Medium	SQLite	AES-256	Private reflections on stage performance
Laugh scores	Medium	SQLite	AES-256	Performance history data
User profile	Medium	AsyncStorage	None (Phase 1)	Display name, comedy style — encrypted in Phase 2
Idea content	High	SQLite	AES-256	Raw, unpolished creative thoughts — most private field
App settings	Low	AsyncStorage	None	Non-sensitive preferences

10.2 Phase 1 Privacy Commitments

- **Zero network calls:** No outbound requests. No analytics, crash reporting, or telemetry in Phase 1.
- **No third-party SDKs:** Only Expo-managed, open-source packages — every dependency is auditable.
- **Creative ownership:** All material stays on-device. Uninstalling permanently deletes all data.
- **No login required:** Phase 1 requires no account. The user owns their data entirely.
- **Voice data:** Voice recordings never leave the device. Speech-to-text uses on-device APIs only.
- **Data export:** Settings → Export Data produces a ZIP containing a JSON file of all data + recordings folder.

10.3 Phase 2 GDPR & Privacy Controls

Right / Control	Implementation
Right of Access	Settings → Export Data: full ZIP of JSON data + recording files, downloadable to device
Right to Erasure	Settings → Delete Account: Edge Function hard-deletes all rows, recordings, and auth user
Data Minimisation	Email + display name only at signup. No phone, no location, no device identifiers stored
Consent	Explicit T&C; + Privacy Policy checkbox at signup; timestamp stored to DB
Creative IP	T&C; explicitly states: all content belongs to the user; Anthropic/company holds zero rights to joke material
Data Residency	Supabase project in ap-south-1 (Mumbai) for Indian user compliance
Row Level Security	All Supabase tables: USING (auth.uid() = user_id). Users access only their own rows

11. TESTING PROTOCOLS & QUALITY ASSURANCE

Minimum **80% line coverage** enforced in CI. The Setlist time calculation, joke version pruning, and all data integrity rules require **95% coverage**. Tests run on every pull request before merge is permitted.

11.1 Test Case Matrix

Module	ID	Scenario	Expected Result	Pri	Phase
Idea Inbox	IDX-01	Capture idea via text	Saved to SQLite; status=Raw; searchable	P0	MVP
Idea Inbox	IDX-02	Record voice clip	Audio file saved; path in idea row	P0	MVP
Idea Inbox	IDX-03	Promote idea to Developing	Status updated; badge color changes	P0	MVP
Joke Builder	JKB-01	Create joke with Setup only	Joke saved as Draft; no validation error	P0	MVP
Joke Builder	JKB-02	Add to setlist without punchline	Blocked; 'Stage Ready' state required	P0	MVP
Joke Builder	JKB-03	Edit joke — version snapshot created	New row in joke_versions; version_number increments	P0	MVP
Joke Builder	JKB-04	Edit 21st time — oldest version pruned	MAX 20 rows enforced; oldest deleted	P0	MVP
Joke Builder	JKB-05	Revert to version 3	Current joke fields = v3 snapshot; new version logged	P0	v1.1
Setlist	SLT-01	Add joke to setlist	Joke appears; order_index assigned; total time updates	P0	MVP
Setlist	SLT-02	Reorder jokes by drag	order_index values rewritten; persists on restart	P0	MVP
Setlist	SLT-03	Total time exceeds 60 min	Warning shown; add blocked	P0	MVP
Setlist	SLT-04	Total time calculation — 3 jokes	SUM(durations) displayed correctly	P0	MVP
Practice	PRC-01	Timer starts and counts down	Timer runs accurately; ring animates	P0	MVP
Practice	PRC-02	Pause then resume	Timer freezes on pause; continues correctly on resume	P0	MVP
Practice	PRC-03	Skip to next joke	Correct joke shown; timer adjusted	P1	MVP
Ratings	RAT-01	Rate joke after practice session	Row created in joke_ratings; avg_score updated	P0	v1.1
Ratings	RAT-02	Rate same joke twice same date	Second rating updates existing row	P0	v1.1
Versions	VER-01	View version history	All versions listed; dates shown	P1	v1.1
Data	DAT-01	Export all data	Valid ZIP produced; JSON parseable	P1	v1.1
Data	DAT-02	Reset all data	All SQLite + AsyncStorage + recordings cleared	P0	v1.1

12. CI/CD PIPELINE, BUILD & DEPLOYMENT

12.1 CI/CD Pipeline

Step	Action	Tool	Trigger	On Failure
1	Lint + Format	ESLint/Prettier	Every PR	Block merge — zero lint errors allowed
2	Type Check	tsc --noEmit	Every PR	Block merge — no TypeScript errors
3	Unit Tests	Jest	Every PR	Block if coverage < 80% or any test fails
4	E2E Tests	Maestro	Pre-release	Block release — joke flow and setlist flow must pass
5	Preview Build	EAS Build	Merge to main	Notify Slack; non-blocking
6	Internal Beta	EAS Submit	Weekly	TestFlight internal / Play Internal Track
7	Production Release	EAS Build+Submit	Manual gate	Human approval + QA sign-off required

12.2 Phased Deployment Plan

Phase	Milestone	Scope	Target	Platform
Alpha	Internal dev	Idea Inbox, Joke Builder (no AI), Setlist, Practice Timer	Week 4	iOS+Android
Beta	TestFlight/Play	Full MVP + Ratings, Version Control, Performance Recorder, Export	Week 8	iOS+Android
v1.0	App Store launch	Complete offline MVP — no account required, full feature set	Week 12	iOS+Android
v1.1	Update	Version Control diff view, Settings, Share Sheet extension	Week 16	iOS+Android
v2.0	Major update	Supabase auth + sync, AI (Punchlines, Expander, Laugh Analyzer)	Week 24	iOS+Android
v2.1	Feature update	Open Mic Finder (Eventbrite API), Topic Explorer, Character Builder	Week 28	iOS+Android

12.3 Performance Targets

Metric	Target	Measurement Method
App cold start	< 2.0 sec	Expo Dev Tools startup profiler
Idea capture screen open	< 300 ms	From tap to input-ready; React Native Perf Monitor
Setlist total time update	< 50 ms	Jest timing test; recomputed on every drag
SQLite write (1 row)	< 10 ms	Jest timing assertions in unit tests
Joke Builder auto-save	< 200 ms	Time from inactivity to SQLite confirmation
Drag reorder haptic	< 16 ms	Reanimated frame profiler; 60fps target
JS bundle size (gzip)	< 3 MB	Metro bundle analyser — audio/video libs add size
RAM usage (idle)	< 120 MB	Android Profiler / Xcode Instruments

13. UI COMPONENT LIBRARY & ANIMATION SPECIFICATIONS

All components are custom-built — no third-party UI library. This ensures full control of the dark OLED + gold design language and eliminates conflicting default styles.

13.1 Core Component Specifications

Component	Key Props	Specification
GoldCTA	label, onPress, variant	#FFB800 primary; #9B72CF secondary; 24dp radius; 48dp height; dark text on gold, white on purple
IdeaCard	idea, onPress, onLongPress	Dark card #252535; status badge top-right (color-coded); title bold white; preview text gray
JokeNotepad	setup, punchline, tag, onChange	#FFF8E1 cream surface; ruled lines; spiral binding left; purple punchline label; auto-save indicator
SetlistItem	joke, index, duration, onDrag, onDelete	Dark card; numbered gold index; drag handle right; duration gray right-aligned; gold for 'Indian Parents' style highlight
TimerRing	totalSecs, remainingSecs, size	SVG arc; gold fill; shrinks as time depletes; changes amber then red at 25% and 10%
StatusBadge	status: IdeaStatus	Pill shape; color-coded: Raw=red, Developing=amber, Stage Ready=green; uppercase 9sp text
ProgressBarsChart	data: DayData[], days	Alternating gold/purple bars; M-I day labels; gold = odd days; purple = even; no axes needed
QuickActionButton	icon, label, onPress	Dark square card #252535; gold icon center; gray label below; 3-column grid layout
DailyGoalWidget	goalText, progressPct	Dark card; bold white goal text; gold time highlight; dual-color donut ring top-right
VersionTimeline	versions: JokeVersion[]	Horizontal scroll; gold dot per version; version number + date below; active = larger dot
TagChip	label, onRemove	#E5DCC8 cream bg inside notepad; #252535 bg on dark screens; close icon on removable chips
TeleprompterView	jokes: Joke[], currentIndex	22sp bold white text; center-aligned; current joke name gold top; fade transition between jokes

13.2 Animation Specifications

Animation	Library	Duration	Easing	Trigger
Timer ring depletion	Reanimated + SVG	Continuous	linear	Every second of practice timer
Timer ring color shift	Reanimated	500ms	ease-in-out	At 25% and 10% remaining time
Drag reorder	Reanimated	Continuous	spring	User drag gesture in Setlist Builder
Card entrance	Reanimated FadeIn	250ms	ease-in	FlatList item mount
Status badge change	Reanimated	300ms	ease-out	Status transition on any idea
Gold CTA press	React Native	100ms	ease-in-out	Scale 0.97 on press-in; 1.0 on press-out
Bottom nav active	Reanimated	200ms	spring	Tab switch — icon scales up + color transitions
Notepad page flip	Reanimated 3D	400ms	ease-in-out	Switching between joke versions in version view

14. APPENDIX — GLOSSARY, ROADMAP & DEVELOPER NOTES

14.1 Glossary

Term	Definition
Setup	The premise of a joke — the observational statement that creates expectation in the audience
Punchline	The twist, subversion, or reveal that delivers the laugh — the payoff of the Setup
Tag	A second, shorter punchline that rides the wave of the first laugh; extends the bit
Callback	A reference back to an earlier joke in the set; rewards attentive audiences
Bit	A complete joke unit: Setup + one or more Punchlines; may span several minutes
Setlist	An ordered sequence of bits selected for a specific performance set with timing
Open Mic	An informal performance venue where comedians test new material before paid gigs
Crowd Work	Spontaneous interaction with the audience; cannot be scripted but can be noted
Laugh Score	User-assigned 1–10 rating of audience reaction to a specific joke at a specific show
Version Control	Automatic snapshot of joke content on every edit; allows reverting to prior drafts
Premise	The core observation or idea behind a joke before it has been shaped into Setup/Punchline
Comedy Style	The user's primary performance mode: Observational / Dark / Storytelling / Roast / Self-dep. / Absurdist
Crowd Type	The audience demographic a joke or setlist is optimised for: Corporate / College / Club / Any
Idea Inbox	The rapid-capture screen; first stop for all raw comedy ideas before structuring
Premise Expander	AI tool (Phase 2) that generates multiple angles from a single joke premise
Punchline Finder	AI tool (Phase 2) that generates candidate punchlines from a user-written Setup
Laugh Analyzer	AI tool (Phase 2) that detects laughter intensity in a recorded performance audio file
Comedy OS	The product's vision statement: a complete operating system for stand-up comedy creation
Phase 1	Offline-only MVP — no internet, no account, all data on-device
Phase 2	Cloud-connected version: Supabase auth, sync, AI features, Open Mic Finder

14.2 Future Roadmap

Ver.	Feature	Description
v2.1	Open Mic Finder	Map-based discovery of nearby open mic events using Eventbrite / Meetup APIs + location permission
v2.1	Topic Explorer	Writing prompt generator with 25+ categories; AI-expanded prompts using comedy style profile
v2.2	Character Builder	Design recurring characters for storytelling sets; link characters to specific bits
v2.2	Laugh Analyzer	Upload performance audio; ML model identifies laugh clusters and intensities per joke
v3.0	Comedy Community	Opt-in sharing of non-identifying joke metrics; upvote promising premises from other comedians
v3.0	Coach Mode	Mentor or comedy teacher can view student's setlists and leave timestamped comments on recordings
v3.1	Watch Companion	Apple Watch / WearOS complication: current joke name + timer remaining during live performance
v3.1	Venue Intelligence	Track performance history by venue; 'what worked at this club last time' recommendations

14.3 Developer Quick-Start Notes

- Run **npm install** then **npx expo start**. Scan QR code with Expo Go on your phone.
- SQLite DB auto-created on first launch at *FileSystem.documentDirectory* + *'standup.db'*.
- All SQL migrations in *src/db/migrations/* — run in sequence on every app startup.
- Voice recordings stored at *FileSystem.documentDirectory* + *'recordings/'* — ensure directory exists on launch.
- Setlist total time is a computed field — **never store it as a primary source of truth**; always recompute from joke durations.
- Version pruning logic lives in *src/features/jokes/versionRepository.ts* — call after every joke save.
- Never store auth tokens in AsyncStorage — Expo SecureStore is mandatory for any sensitive values in Phase 2.
- Follow FSD architecture strictly: no cross-feature imports except through the *src/shared/* layer.
- **Environment variables:** Copy *.env.example* to *.env*. Phase 1 requires no values. Phase 2 needs SUPABASE_URL and SUPABASE_ANON_KEY.